



PROBABILITY THEORY

MTM2592-Gr:2

SIXTH WEEK

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CONDITIONAL DISTRIBUTION:

In the conditional probability of event A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) \neq 0$. Suppose now that A and B are the events $X = x$ and $Y = y$ so that we can write

$$\begin{aligned} P(X = x|Y = y) &= \frac{P(X = x, Y = y)}{P(Y = y)} \\ &= \frac{f(x, y)}{h(y)} \end{aligned}$$

provided $P(Y = y) = h(y) \neq 0$, where $f(x, y)$ is the value of the joint probability distribution of X and Y at (x, y) , and $h(y)$ is the value of the marginal distribution of Y at y . Denoting the conditional probability by $f(x|y)$ to indicate that x is a variable and y is fixed, let us now make the following definition.

DEFINITION **CONDITIONAL DISTRIBUTION.** *If $f(x, y)$ is the value of the joint probability distribution of the discrete random variables X and Y at (x, y) and $h(y)$ is the value of the marginal distribution of Y at y , the function given by*

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h(y) \neq 0$$

for each x within the range of X is called the conditional distribution of X given $Y = y$. Correspondingly, if $g(x)$ is the value of the marginal distribution of X at x , the function given by

$$w(y|x) = \frac{f(x, y)}{g(x)} \quad g(x) \neq 0$$

for each y within the range of Y is called the conditional distribution of Y given $X = x$.

EXAMPLE
given $Y = 1$.

Consider the caplet example, find the conditional distribution of X

Solution

Substituting the appropriate values from the table in caplet example, we find

$$f(0|1) = \frac{\frac{2}{9}}{\frac{7}{18}} = \frac{4}{7}$$

$$f(1|1) = \frac{\frac{1}{6}}{\frac{7}{18}} = \frac{3}{7}$$

$$f(2|1) = \frac{0}{\frac{7}{18}} = 0$$

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

DEFINITION **CONDITIONAL DENSITY.** *If $f(x, y)$ is the value of the joint density of the continuous random variables X and Y at (x, y) and $h(y)$ is the value of the marginal distribution of Y at y , the function given by*

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h(y) \neq 0$$

for $-\infty < x < \infty$, is called the conditional density of X given $Y = y$. Correspondingly, if $g(x)$ is the value of the marginal density of X at x , the function given by

$$w(y|x) = \frac{f(x, y)}{g(x)} \quad g(x) \neq 0$$

for $-\infty < y < \infty$, is called the conditional density of Y given $X = x$.

EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the conditional density of X given $Y = y$, and

use it to evaluate $P(X \leq \frac{1}{2} | Y = \frac{1}{2})$.

Solution

$$\begin{aligned} f(x|y) &= \frac{f(x, y)}{h(y)} = \frac{\frac{2}{3}(x + 2y)}{\frac{1}{3}(1 + 4y)} \\ &= \frac{2x + 4y}{1 + 4y} \end{aligned}$$

for $0 < x < 1$ and $f(x|y) = 0$ elsewhere. Now,

and we can write

$$\begin{aligned} f\left(x \middle| \frac{1}{2}\right) &= \frac{2x + 4 \cdot \frac{1}{2}}{1 + 4 \cdot \frac{1}{2}} \\ &= \frac{2x + 2}{3} \end{aligned} \quad \Rightarrow \quad P\left(X \leq \frac{1}{2} \middle| Y = \frac{1}{2}\right) = \int_0^{\frac{1}{2}} \frac{2x + 2}{3} dx = \frac{5}{12}$$

EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} 4xy & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the marginal densities of X and Y and the conditional density of X given $Y = y$.

Solution

Performing the necessary integrations, we get

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 4xy dy \\ &= 2xy^2 \Big|_{y=0}^{y=1} = 2x \end{aligned}$$

for $0 < x < 1$, and $g(x) = 0$ elsewhere; also

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 4xy dx \\ &= 2x^2y \Big|_{x=0}^{x=1} = 2y \end{aligned}$$

for $0 < y < 1$, and $h(y) = 0$ elsewhere. Then, substituting into the formula for a conditional density, we get

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{4xy}{2y} = 2x$$

for $0 < x < 1$, and $f(x|y) = 0$ elsewhere.

The Multivariate Case

When we are dealing with more than two random variables, whether continuous or discrete, we can consider various different kinds of conditional distributions or densities. For instance, if $f(x_1, x_2, x_3, x_4)$ is the value of the joint distribution of the discrete random variables X_1, X_2, X_3 , and X_4 at (x_1, x_2, x_3, x_4) , we can write

$$p(x_3|x_1, x_2, x_4) = \frac{f(x_1, x_2, x_3, x_4)}{g(x_1, x_2, x_4)} \quad g(x_1, x_2, x_4) \neq 0$$

for the value of the conditional distribution of X_3 at x_3 given $X_1 = x_1, X_2 = x_2$, and $X_4 = x_4$, where $g(x_1, x_2, x_4)$ is the value of the joint marginal distribution of X_1, X_2 , and X_4 at (x_1, x_2, x_4) . We can also write

$$q(x_2, x_4|x_1, x_3) = \frac{f(x_1, x_2, x_3, x_4)}{m(x_1, x_3)} \quad m(x_1, x_3) \neq 0$$

for the value of the **joint conditional distribution** of X_2 and X_4 at (x_2, x_4) given $X_1 = x_1$ and $X_3 = x_3$, or

$$r(x_2, x_3, x_4|x_1) = \frac{f(x_1, x_2, x_3, x_4)}{b(x_1)} \quad b(x_1) \neq 0$$

for the value of the joint conditional distribution of X_2, X_3 , and X_4 at (x_2, x_3, x_4) given $X_1 = x_1$.

Ex:

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3}, & 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0, & \text{elsewhere} \end{cases}$$

find $f(x_2 | x_1, x_3)$ and $P\left(x_2 \leq \frac{1}{2} \middle| x_1 = \frac{1}{2}, x_3 = 1\right)$

Solution:

$$f(x_2 | x_1, x_3) = \frac{f(x_1, x_2, x_3)}{m(x_1, x_3)} = \frac{(x_1 + x_2)e^{-x_3}}{\left(x_1 + \frac{1}{2}\right)e^{-x_3}} = \frac{x_1 + x_2}{x_1 + \frac{1}{2}} \quad \text{for } 0 < x_2 < 1$$

$$f(x_2 | x_1 = \frac{1}{2}, x_3 = 1) = \frac{x_1 + x_2}{x_1 + \frac{1}{2}} = \frac{\frac{1}{2} + x_2}{1}$$

$$P\left(x_2 \leq \frac{1}{2} \middle| x_1 = \frac{1}{2}, x_3 = 1\right) = \int_0^{1/2} \left(\frac{1}{2} + x_2\right) dx_2 = \frac{3}{8}$$

INDEPENDENCE

DEFINITION **INDEPENDENCE OF DISCRETE RANDOM VARIABLES.** If $f(x_1, x_2, \dots, x_n)$ is the value of the joint probability distribution of the discrete random variables $X_1, X_2, \dots, X_n$ at $(x_1, x_2, \dots, x_n)$ and $f_i(x_i)$ is the value of the marginal distribution of X_i at x_i for $i = 1, 2, \dots, n$, then the n random variables are independent if and only if

$$f(x_1, x_2, \dots, x_n) = f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n)$$

for all $(x_1, x_2, \dots, x_n)$ within their range.

Theorem: If random variables X and Y are independent. Conditional probability functions are equal to marginal probability functions

$$\begin{aligned} P(x|y) &= P(x), \quad P(y) \neq 0 \\ P(y|x) &= P(y), \quad P(x) \neq 0 \end{aligned}$$

Proof: If random variables X and Y are independent
 $P(x,y)=P(x)P(y)$

$$P(x|y) = \frac{P(x, y)}{P(y)} = \frac{P(x)P(y)}{P(y)} = P(x)$$

EXAMPLE

Considering n independent flips of a balanced coin, let X_i be the number of heads (0 or 1) obtained in the i th flip for $i = 1, 2, \dots, n$. Find the joint probability distribution of these n random variables.

Solution

Since each of the random variables X_i , for $i = 1, 2, \dots, n$, has the probability distribution

$$f_i(x_i) = \frac{1}{2} \quad \text{for } x_i = 0, 1$$

and the n random variables are independent, their joint probability distribution is given by

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \dots \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^n \end{aligned}$$

where $x_i = 0$ or 1 for $i = 1, 2, \dots, n$.

EXAMPLE

Given the independent random variables X_1 , X_2 , and X_3 with the probability densities

$$f_1(x_1) = \begin{cases} e^{-x_1} & \text{for } x_1 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_2(x_2) = \begin{cases} 2e^{-2x_2} & \text{for } x_2 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_3(x_3) = \begin{cases} 3e^{-3x_3} & \text{for } x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find their joint probability density, and use it to evaluate the probability $P(X_1 + X_2 \leq 1, X_3 > 1)$.

Solution

$$\begin{aligned} f(x_1, x_2, x_3) &= f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3) \\ &= e^{-x_1} \cdot 2e^{-2x_2} \cdot 3e^{-3x_3} \\ &= 6e^{-x_1-2x_2-3x_3} \end{aligned}$$

for $x_1 > 0, x_2 > 0, x_3 > 0$, and $f(x_1, x_2, x_3) = 0$ elsewhere. Thus,

$$\begin{aligned} P(X_1 + X_2 \leq 1, X_3 > 1) &= \int_1^\infty \int_0^1 \int_0^{1-x_2} 6e^{-x_1-2x_2-3x_3} dx_1 dx_2 dx_3 \\ &= (1 - 2e^{-1} + e^{-2})e^{-3} \\ &= 0.020 \end{aligned}$$

THE EXPECTED VALUE FOR MULTIVARIATE CASE

THEOREM If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , the expected value of $g(X, Y)$ is

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot f(x, y)$$

Correspondingly, if X and Y are continuous random variables and $f(x, y)$ is the value of their joint probability density at (x, y) , the expected value of $g(X, Y)$ is

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

EXAMPLE

Consider the caplets example. Find the expected value of $g(X, Y) = X + Y$.

Solution

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

$$\begin{aligned} E(X + Y) &= \sum_{x=0}^2 \sum_{y=0}^2 (x + y) \cdot f(x, y) \\ &= (0 + 0) \cdot \frac{1}{6} + (0 + 1) \cdot \frac{2}{9} + (0 + 2) \cdot \frac{1}{36} + (1 + 0) \cdot \frac{1}{3} \\ &\quad + (1 + 1) \cdot \frac{1}{6} + (2 + 0) \cdot \frac{1}{12} \\ &= \frac{10}{9} \end{aligned}$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \cdot f(x, y) \, dx \, dy$$

$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xf(x, y) \, dx \, dy = \int_{-\infty}^{+\infty} xf(x) \, dx = \mu_x$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} yf(x, y) \, dx \, dy = \int_{-\infty}^{+\infty} yf(y) \, dy = \mu_y$$

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{2}{7}(x + 2y) & \text{for } 0 < x < 1, 1 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X, Y) = X/Y^3$.

Solution

$$\begin{aligned} E(X/Y^3) &= \int_1^2 \int_0^1 \frac{2x(x + 2y)}{7y^3} dx dy \\ &= \frac{2}{7} \int_1^2 \left(\frac{1}{3y^3} + \frac{1}{y^2} \right) dy \\ &= \frac{15}{84} \end{aligned}$$

THEOREM If $c_1, c_2, \dots$, and c_n are constants, then

$$E \left[\sum_{i=1}^n c_i g_i(X_1, X_2, \dots, X_k) \right] = \sum_{i=1}^n c_i E[g_i(X_1, X_2, \dots, X_k)]$$

PRODUCT MOMENTS

DEFINITION **PRODUCT MOMENTS ABOUT THE ORIGIN.** *The r th and s th product moment about the origin of the random variables X and Y , denoted by $\mu'_{r,s}$, is the expected value of $X^r Y^s$; symbolically,*

$$\mu'_{r,s} = E(X^r Y^s) = \sum_x \sum_y x^r y^s \cdot f(x, y)$$

for $r = 0, 1, 2, \dots$ and $s = 0, 1, 2, \dots$ when X and Y are discrete, and

$$\mu'_{r,s} = E(X^r Y^s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^r y^s \cdot f(x, y) dx dy$$

when X and Y are continuous.

DEFINITION **PRODUCT MOMENTS ABOUT THE MEAN.** *The r th and s th product moment about the means of the random variables X and Y , denoted by $\mu_{r,s}$, is the expected value of $(X - \mu_X)^r (Y - \mu_Y)^s$; symbolically,*

$$\begin{aligned}\mu_{r,s} &= E[(X - \mu_X)^r (Y - \mu_Y)^s] \\ &= \sum_x \sum_y (x - \mu_X)^r (y - \mu_Y)^s \cdot f(x, y)\end{aligned}$$

for $r = 0, 1, 2, \dots$ and $s = 0, 1, 2, \dots$ when X and Y are discrete, and

$$\begin{aligned}\mu_{r,s} &= E[(X - \mu_X)^r (Y - \mu_Y)^s] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^r (y - \mu_Y)^s \cdot f(x, y) dx dy\end{aligned}$$

when X and Y are continuous.

COVARIANCE

DEFINITION COVARIANCE. $\mu_{1,1}$ is called the covariance of X and Y , and it is denoted by σ_{XY} , $\text{cov}(X, Y)$, or $C(X, Y)$.

THEOREM

$$\sigma_{XY} = \mu'_{1,1} - \mu_X \mu_Y$$

Proof Using the various theorems about expected values, we can write

$$\begin{aligned}\sigma_{XY} &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X\mu_Y \\ &= E(XY) - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y \\ &= \mu'_{1,1} - \mu_X\mu_Y\end{aligned}$$

$$\text{Cov}(X, Y) = E(X - E(X))(Y - E(Y)) = E(XY) - E(X)E(Y)$$

$$= E(XY) - \mu_x \mu_y$$

The covariance has the following properties:

1. $\text{Cov}(X, X) = \text{Var}(X)$;
2. if X and Y are independent then $\text{Cov}(X, Y) = 0$;
3. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
4. $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$;
5. $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$;
6. $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$;
7. more generally,

$$\text{Cov} \left(\sum_{i=1}^m a_i X_i, \sum_{j=1}^n b_j Y_j \right) = \sum_{i=1}^m \sum_{j=1}^n a_i b_j \text{Cov}(X_i, Y_j).$$

EXAMPLE

Consider the caplet example.

The joint and marginal probabilities of X and Y , the numbers of aspirin and sedative caplets among two caplets drawn at random from a bottle containing three aspirin, two sedative, and four laxative caplets, are recorded as follows:

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

Find the covariance of X and Y .

Solution

Referring to the joint probabilities given here, we get

$$\begin{aligned}\mu'_{1,1} &= E(XY) \\ &= 0 \cdot 0 \cdot \frac{1}{6} + 0 \cdot 1 \cdot \frac{2}{9} + 0 \cdot 2 \cdot \frac{1}{36} + 1 \cdot 0 \cdot \frac{1}{3} + 1 \cdot 1 \cdot \frac{1}{6} + 2 \cdot 0 \cdot \frac{1}{12} \\ &= \frac{1}{6}\end{aligned}$$

and using the marginal probabilities, we get

$$\mu_X = E(X) = 0 \cdot \frac{5}{12} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{12} = \frac{2}{3}$$

and

$$\mu_Y = E(Y) = 0 \cdot \frac{7}{12} + 1 \cdot \frac{7}{18} + 2 \cdot \frac{1}{36} = \frac{4}{9}$$

It follows that

$$\sigma_{XY} = \frac{1}{6} - \frac{2}{3} \cdot \frac{4}{9} = -\frac{7}{54}$$

The negative result suggests that the more aspirin tablets we get the fewer sedative tablets we will get, and vice versa, and this, of course, makes sense.

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

EXAMPLE

Find the covariance of the random variables whose joint probability density is given by

$$f(x, y) = \begin{cases} 2 & \text{for } x > 0, y > 0, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Solution

Evaluating the necessary integrals, we get

$$E(X) = \mu_X = \int_0^1 \int_0^{1-x} 2x \, dy \, dx = \frac{1}{3}$$

$$E(Y) = \mu_Y = \int_0^1 \int_0^{1-x} 2y \, dy \, dx = \frac{1}{3}$$

and

$$E(XY) = \int_0^1 \int_0^{1-x} 2xy \, dy \, dx = \frac{1}{12}$$

It follows that

$$\sigma_{XY} = \frac{1}{12} - \frac{1}{3} \cdot \frac{1}{3} = -\frac{1}{36}$$

THEOREM If X and Y are independent, then $E(XY) = E(X) \cdot E(Y)$ and $\sigma_{XY} = 0$.

Proof For the discrete case we have, by definition,

$$E(XY) = \sum_x \sum_y xy \cdot f(x, y)$$

Since X and Y are independent, we can write $f(x, y) = g(x) \cdot h(y)$, where $g(x)$ and $h(y)$ are the values of the marginal distributions of X and Y , and we get

$$\begin{aligned} E(XY) &= \sum_x \sum_y xy \cdot g(x)h(y) \\ &= \left[\sum_x x \cdot g(x) \right] \left[\sum_y y \cdot h(y) \right] \\ &= E(X) \cdot E(Y) \end{aligned}$$

Hence,

$$\begin{aligned} \sigma_{XY} &= \mu'_{1,1} - \mu_X \mu_Y \\ &= E(X) \cdot E(Y) - E(X) \cdot E(Y) \\ &= 0 \end{aligned}$$

Note that the independency of two variables implies a zero covariance, but a zero covariance does not necessarily imply their independence.

EXAMPLE

If the joint probability distribution of X and Y is given by show that their covariance is zero even though the two random variables are not independent.

Solution

Using the probabilities shown in the margins, we get

$$\mu_X = (-1) \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0$$

$$\mu_Y = (-1) \cdot \frac{2}{3} + 0 \cdot 0 + 1 \cdot \frac{1}{3} = -\frac{1}{3}$$

and

$$\begin{aligned} E(XY) &= \mu'_{1,1} = (-1)(-1) \cdot \frac{1}{6} + 0(-1) \cdot \frac{1}{3} + 1(-1) \cdot \frac{1}{6} + (-1)1 \cdot \frac{1}{6} + 1 \cdot 1 \cdot \frac{1}{6} \\ &= 0 \end{aligned}$$

Thus, $\sigma_{XY} = 0 - 0(-\frac{1}{3}) = 0$, the covariance is zero, but the two random variables are not independent. For instance, $f(x, y) \neq g(x) \cdot h(y)$ for $x = -1$ and $y = -1$.

	x			
	-1	0	1	
-1	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{2}{3}$
0	0	0	0	0
1	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	

THEOREM If $X_1, X_2, \dots, X_n$ are independent, then

$$E(X_1 X_2 \cdots X_n) = E(X_1) \cdot E(X_2) \cdots E(X_n)$$

Definition: Variance of the sum of the two random variables X and Y

$$\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)+2\text{Cov}(X,Y)$$

Variance of the difference of two random variables,

$$\text{Var}(X-Y)=\text{Var}(X)+\text{Var}(Y)-2\text{Cov}(X,Y)$$

CORRELATION

Definition: Let μ_x and μ_y be means of X and Y and σ_x and σ_y be standard deviation of X and Y, respectively. The correlation of X and Y is defined as;

$$\text{corr}(X, Y) = \rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sqrt{E[(X - \mu_x)^2]} \sqrt{E[(Y - \mu_y)^2]}}$$

$\text{corr}(X,Y)$ is dimensionless, while $\text{cov}(X,Y)$ has units.

Correlation coefficient; $-1 \leq \rho_{xy} \leq 1$

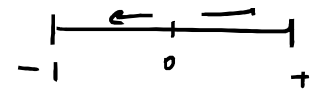
$\rho_{xy} = 0$ There is no linear correlation

$\rho_{xy} < 0$ inverse linear correlation ✓

$\rho_{xy} > 0$ linear correlation ✓

$\rho_{xy} = -1$ There is a perfect inverse linear correlation

$\rho_{xy} = +1$ There is a perfect linear correlation



The degree of the relationship gets stronger as it move away from zero.

Theorem: If X and Y random variables provide $Y = aX + b$ equation

$$\rho_{xy} = \begin{cases} -1, & a < 0 \\ 0, & a = 0 \\ 1, & a > 0 \end{cases}$$

$$\rho_{xy} = \rho_{yx}$$

MOMENTS OF LINEAR COMBINATIONS OF RANDOM VARIABLES

THEOREM If $X_1, X_2, \dots, X_n$ are random variables and

$$Y = \sum_{i=1}^n a_i X_i$$

where $a_1, a_2, \dots, a_n$ are constants, then

$$E(Y) = \sum_{i=1}^n a_i E(X_i)$$

and

$$\text{var}(Y) = \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i) + 2 \sum_{i < j} a_i a_j \cdot \text{cov}(X_i, X_j)$$

where the double summation extends over all values of i and j , from 1 to n , for which $i < j$.

Proof

$$E(Y) = E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$$

$$\begin{aligned} \text{var}(Y) &= E\left([Y - E(Y)]^2\right) = E\left\{\left[\sum_{i=1}^n a_i X_i - \sum_{i=1}^n a_i E(X_i)\right]^2\right\} \\ &= E\left\{\left[\sum_{i=1}^n a_i (X_i - \mu_i)\right]^2\right\} \end{aligned}$$

Then, expanding by means of the multinomial theorem, according to which $(a + b + c + d)^2$, for example, equals $a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd$, and again referring to Theorem we get

$$\begin{aligned}\text{var}(Y) &= \sum_{i=1}^n a_i^2 E[(X_i - \mu_i)^2] + 2 \sum_{i < j} a_i a_j E[(X_i - \mu_i)(X_j - \mu_j)] \\ &= \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i) + 2 \sum_{i < j} a_i a_j \cdot \text{cov}(X_i, X_j)\end{aligned}$$

Note that we have tacitly made use of the fact that $\text{cov}(X_i, X_j) = \text{cov}(X_j, X_i)$.

Since $\text{cov}(X_i, X_j) = 0$ when X_i and X_j are independent, we obtain the following corollary.

COROLLARY If the random variables $X_1, X_2, \dots, X_n$ are independent and

$Y = \sum_{i=1}^n a_i X_i$, then

$$\text{var}(Y) = \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i)$$

EXAMPLE

If the random variables X , Y , and Z have the means $\mu_X = 2$, $\mu_Y = -3$, and $\mu_Z = 4$, the variances $\sigma_X^2 = 1$, $\sigma_Y^2 = 5$, and $\sigma_Z^2 = 2$, and the covariances $\text{cov}(X, Y) = -2$, $\text{cov}(X, Z) = -1$, and $\text{cov}(Y, Z) = 1$, find the mean and the variance of $W = 3X - Y + 2Z$.

Solution

$$\begin{aligned} E(W) &= E(3X - Y + 2Z) \\ &= 3E(X) - E(Y) + 2E(Z) \\ &= 3 \cdot 2 - (-3) + 2 \cdot 4 \\ &= 17 \end{aligned}$$

and

$$\begin{aligned} \text{var}(W) &= 9 \text{ var}(X) + \text{var}(Y) + 4 \text{ var}(Z) - 6 \text{ cov}(X, Y) \\ &\quad + 12 \text{ cov}(X, Z) - 4 \text{ cov}(Y, Z) \\ &= 9 \cdot 1 + 5 + 4 \cdot 2 - 6(-2) + 12(-1) - 4 \cdot 1 \\ &= 18 \end{aligned}$$

THEOREM If $X_1, X_2, \dots, X_n$ are random variables and

$$Y_1 = \sum_{i=1}^n a_i X_i \quad \text{and} \quad Y_2 = \sum_{i=1}^n b_i X_i$$

where $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are constants, then

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^n a_i b_i \cdot \text{var}(X_i) + \sum_{i < j} (a_i b_j + a_j b_i) \cdot \text{cov}(X_i, X_j)$$

COROLLARY If the random variables $X_1, X_2, \dots, X_n$ are **independent**, $Y_1 = \sum_{i=1}^n a_i X_i$ and $Y_2 = \sum_{i=1}^n b_i X_i$, then

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^n a_i b_i \cdot \text{var}(X_i)$$

Example: If the random variables X, Y and Z have the means $\mu_x = 3, \mu_y = 5, \mu_z = 2$, and variances $\sigma_x^2 = 8, \sigma_y^2 = 12, \sigma_z^2 = 18$ and covariances $Cov(X, Y) = 1, Cov(X, Z) = -3, Cov(Y, Z) = 2$

i) Find the expected value, variance and covariance of

$$U = X - 4Y + 2Z \text{ and } V = 3X - Y - Z$$

ii) If the random variables X, Y and Z are independent find the expected value, variance and covariance of

$$U = X - 4Y + 2Z \text{ and } V = 3X - Y - Z$$

Solution:

$$U = X - 4Y + 2Z, V = 3X - Y - Z \quad \Rightarrow \quad X_1 = X, X_2 = Y, X_3 = Z; a_1 = 1, a_2 = -4, a_3 = 2; b_1 = 3, b_2 = -1, b_3 = -1$$

$$U = a_1X_1 + a_2X_2 + a_3X_3, V = b_1X_1 + b_2X_2 + b_3X_3$$

$$E(U) = E(a_1X_1 + a_2X_2 + a_3X_3) = a_1E(X_1) + a_2E(X_2) + a_3E(X_3) = 1.3 + (-4).5 + 2.2 = -13$$

$$E(V) = E(b_1X_1 + b_2X_2 + b_3X_3) = b_1E(X_1) + b_2E(X_2) + b_3E(X_3) = 3.3 + (-1).5 + (-1).2 = 2$$

$$\begin{aligned} Var(U) &= Var(a_1X_1 + a_2X_2 + a_3X_3) = a_1^2Var(X_1) + a_2^2Var(X_2) + a_3^2Var(X_3) + 2a_1a_2Cov(X_1, X_2) + 2a_1a_3Cov(X_1, X_3) + 2a_2a_3Cov(X_2, X_3) \\ &= 1^2.8 + (-4)^2.12 + 2^2.18 + 2.1.(-4).1 + 2.1.2.(-3) + 2.(-4).2.2 = 8 + 192 + 72 - 8 - 12 - 32 = 220 \end{aligned}$$

$$\begin{aligned} Var(V) &= Var(b_1X_1 + b_2X_2 + b_3X_3) = b_1^2Var(X_1) + b_2^2Var(X_2) + b_3^2Var(X_3) + 2b_1b_2Cov(X_1, X_2) + 2b_1b_3Cov(X_1, X_3) + 2b_2b_3Cov(X_2, X_3) \\ &= 3^2.8 + (-1)^2.12 + (-1)^2.18 + 2.3.(-1).1 + 2.3.(-1).(-3) + 2.(-1).(-1).2 = 72 + 12 + 18 - 6 + 18 + 4 = 192 \end{aligned}$$

$$\begin{aligned} Cov(U, V) &= a_1b_1Var(X_1) + a_2b_2Var(X_2) + a_3b_3Var(X_3) + (a_1b_2 + a_2b_1)Cov(X_1, X_2) + (a_1b_3 + a_3b_1)Cov(X_1, X_3) + (a_2b_3 + a_3b_2)Cov(X_2, X_3) \\ &= 1.3.8 + (-4).(-1).12 + 2.(-1).18 + (1.(-1) + (-4).3).1 + (1.(-1) + 2.3).(-3) + ((-4).(-1) + 2.(-1)).2 \\ &= 24 + 48 - 36 - 13 - 15 + 4 = 12 \end{aligned}$$

If the random variables X, Y and Z are independent, then;

$$E(U) = E(a_1X_1 + a_2X_2 + a_3X_3) = a_1E(X_1) + a_2E(X_2) + a_3E(X_3) = 1.3 + (-4).5 + 2.2 = -13$$

$$E(V) = E(b_1X_1 + b_2X_2 + b_3X_3) = b_1E(X_1) + b_2E(X_2) + b_3E(X_3) = 3.3 + (-1).5 + (-1).2 = 2$$

$$\begin{aligned} Var(U) &= Var(a_1X_1 + a_2X_2 + a_3X_3) = a_1^2Var(X_1) + a_2^2Var(X_2) + a_3^2Var(X_3) \\ &= 1^2.8 + (-4)^2.12 + 2^2.18 = 8 + 192 + 72 = 272 \end{aligned}$$

$$\begin{aligned} Var(V) &= Var(b_1X_1 + b_2X_2 + b_3X_3) = b_1^2Var(X_1) + b_2^2Var(X_2) + b_3^2Var(X_3) \\ &= 3^2.8 + (-1)^2.12 + (-1)^2.18 = 72 + 12 + 18 = 102 \end{aligned}$$

$$\begin{aligned} Cov(U, V) &= a_1b_1Var(X_1) + a_2b_2Var(X_2) + a_3b_3Var(X_3) \\ &= 1.3.8 + (-4).(-1).12 + 2.(-1).18 \\ &= 24 + 48 - 36 = 36 \end{aligned}$$

CONDITIONAL EXPECTATION:

If X and Y are random variables and $f(x|y)$ is the value of the conditional probability distribution given $Y = y$, the conditional expectation of $u(X)$, given $Y = y$ is

$$E(u(X)|y) = \begin{cases} \sum_x u(x) \cdot f(x|y) & \text{If } X \text{ and } Y \text{ are discrete} \\ \int_{-\infty}^{\infty} u(x) \cdot f(x|y) dx & \text{If } X \text{ and } Y \text{ are continuous} \end{cases}$$

If X and Y are random variables and $f(y|x)$ is the value of the conditional probability distribution given $X = x$, the conditional expectation of $v(Y)$, given $X = x$ is

$$E(v(Y)|x) = \begin{cases} \sum_y v(y) \cdot f(y|x) & \text{If } X \text{ and } Y \text{ are discrete} \\ \int_{-\infty}^{\infty} v(y) \cdot f(y|x) dy & \text{If } X \text{ and } Y \text{ are continuous} \end{cases}$$

If we let $u(X) = X$, we obtain conditional mean of the random variable X given $Y = y$ which we denote by $\mu_{x|y}$

$$E(X|y) = \mu_{x|y} = \begin{cases} \sum_x x \cdot f(x|y) & \text{If X and Y are discrete} \\ \int_{-\infty}^{\infty} x \cdot f(x|y) dx & \text{If X and Y are continuous} \end{cases}$$

If we let $v(Y) = Y$ we obtain conditional mean of the random variable Y given $X = x$ which we denote by $\mu_{y|x}$

$$E(Y|x) = \mu_{y|x} = \begin{cases} \sum_y y \cdot f(y|x) & \text{If X and Y are discrete} \\ \int_{-\infty}^{\infty} y \cdot f(y|x) dy & \text{If X and Y are continuous} \end{cases}$$

CONDITIONAL VARIANCE:

Correspondingly, the conditional variance of X , given $Y = y$ is denoted by $\sigma_{x|y}^2$ and obtained by:

$$\sigma_{x|y}^2 = E\left(\left(X - \mu_{x|y}\right)^2 | y\right) = E\left(X^2 | y\right) - \left(\mu_{x|y}\right)^2$$

And the conditional variance of Y , given $X = x$ is denoted by $\sigma_{y|x}^2$ and obtained by:

$$\sigma_{y|x}^2 = E\left(\left(Y - \mu_{y|x}\right)^2 | x\right) = E\left(Y^2 | x\right) - \left(\mu_{y|x}\right)^2$$

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{2}{3}(x+2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the conditional mean and the conditional variance of X given $Y = \frac{1}{2}$.

Solution

For these random variables the conditional density of X given $Y = y$ is

$$f(x|y) = \begin{cases} \frac{2x+4y}{1+4y} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

so that

$$f\left(x\left|\frac{1}{2}\right.\right) = \begin{cases} \frac{2}{3}(x+1) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, $\mu_{X|\frac{1}{2}}$ is given by

$$\begin{aligned} E\left(X\left|\frac{1}{2}\right.\right) &= \int_0^1 \frac{2}{3}x(x+1) dx \\ &= \frac{5}{9} \end{aligned}$$

Next we find

$$\begin{aligned} E\left(X^2\left|\frac{1}{2}\right.\right) &= \int_0^1 \frac{2}{3}x^2(x+1) dx \\ &= \frac{7}{18} \end{aligned}$$

and it follows that

$$\sigma_{X|\frac{1}{2}}^2 = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \frac{13}{162}$$

Example: Consider the caplets example.

- Find the conditional mean of X given $Y=1$.
- Find the conditional mean of Y given $X=1$.

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	